COMPUTATIONAL PHILOLOGY · CORPORA · NEGATIVE RESULTS

Reading Steppe Names in Chinese Transcription

Three corpora, two medieval lexicons, and a measured answer to a fifty-year-old question about how much chance a sound law may tolerate.

Aziz M. Uluğ, PhD
Cortechs.ai · aulug@cortechs.ai, ulug@ieee.org
Preprint. Data and code: github.com/ulug2004/chinese-transcription-channel

ABSTRACT

Chinese histories record hundreds of Inner Asian names (Xiongnu rulers, Turkic titles, tribal designations) in characters chosen for their sound. Two thousand years of change in Chinese pronunciation have left the modern Mandarin readings of those characters unrelated to what the scribes heard. This paper models the transcription process itself as a noisy channel, trains it where both sides are known, and measures what can and cannot be recovered.

We generated the following. **Three transcription corpora**, two of them extracted from sources not previously available as data: 9,329 Chinese–Mongolian pairs from the *Secret History of the Mongols*, 594 Chinese–Uyghur pairs from Ligeti's edition of a Ming glossary, and 1,017 verified Chinese–Sanskrit pairs separated out of a 27,110-entry Buddhist dictionary that mixes phonetic spellings with translations. **Two medieval Turkic lexicons** in machine-readable form, for which we found no downloadable dataset: 6,820 glossed headwords from the Türk Dil Kurumu index to Kāşgarî's *Dīwān Lughāt al-Turk* (1072–77), and 2,110 from Kuun's 1880 edition of the *Codex Cumanicus*. **A working channel model**, which on held-out data reconstructs 31% of Turkic words exactly and 84% within two segments. **One positive reading** that survives every test we can apply: 若鞮, the element closing the formal names of six later chanyu, reads as *Inakt* "the faithful ones", since the *Hou Hanshu* glosses it 孝 "filial", the *Codex Cumanicus* glosses *inak* as *fidelis*, and the plural in `-t` is an Old Turkic formation used with titles. **And a result about the most consequential name in the record**: the century-old disagreement over 冒頓, between the standard *Bayatur* and our *Bögü Tın* "sage spirit", turns on which of two Later Han readings of a single character is taken, a choice no published reading of the name states and one the transcription evidence cannot make. And **a set of measured negative results**: language attribution from transcription evidence is not supportable on available reference data, and full reconstruction of the Xiongnu names is not supportable at the density of evidence that survives: 1.4 observations per character, against the 60 that made the same method work on Mongolian.

The measurements also answer the question the field has carried unresolved since 1973. Doerfer argued that chance resemblance is pervasive in long-range comparison; his reviewer sharpened it into a threshold problem: *is it a sound law if 49% of cases show one correspondence and 51% another?* He proposed three classes that nobody could then apply, for want of numbers. Section 7 applies them, with confidence intervals, and finds that the threshold cannot be a constant: it is the baseline of the component being measured, and it moves with how many outcomes that component has.

Keywords: historical phonology · noisy-channel models · corpus construction · Xiongnu · Old Turkic · null models · negative results

1 The problem

When a Chinese scribe of the Han dynasty wrote down a foreign name, he chose the characters for their sound. 冒頓, the founder of the Xiongnu state, is read *Mòdùn* in modern Mandarin, but Mandarin is not the language the characters were chosen in. In the Later Han period those two characters were pronounced approximately as `mək tuənᶜ`.[22] Every argument about what such a name meant, and about who the people bearing it were, depends on getting from the modern reading back to the period one.

That is a recoverable step: reconstructions of Later Han Chinese are published and are not in dispute here. The harder question is the one after it. Given a period reading, can the original foreign word be recovered? The transcription is lossy: Chinese had no `r-`, a restricted set of syllable shapes, and no way to write many foreign sounds. So, the mapping from source word to spelling is many-to-one and its inverse is underdetermined. This is a noisy-channel problem, and it can be treated as one.

The interest of the Xiongnu case is that the answer is unknown and contested: proposals put the ruling house's language in the Turkic, Mongolic, Yeniseian and Iranian families, each resting on a handful of transcribed names; the most recent argues for Yeniseian, specifically an early form of Arin,[3] and Doerfer argued half a century ago that none of the proposals could be sustained on the evidence then available.[6] The interest of this *method* is that its reliability can be measured in cases where the answer is known, which is what most of this paper does.

2 A question the field has already asked

Nothing in the approach below is methodologically new, and it is worth saying so at the outset. Gerhard Doerfer made the argument in 1973, in *Lautgesetz und Zufall* ("Sound Law and Chance"), a study of what he called *omnicomparatism*: relating languages across great distances on the strength of resemblances.[5] Chance produces such resemblances in quantity, was Doerfer's case, and comparativists have a repertoire of devices (he called them tricks) for making weak matches look strong. He included deliberately absurd comparisons of his own to show how easily the devices work.

A review in *Anthropos* the following year pressed the point into a question that can be answered with numbers.[15] Sound laws, the reviewer observed, are not deterministic but probabilistic, and the literature never says how probabilistic they are permitted to be. *Is it a sound law if 49% of cases show the same correspondence and 51% show something else?* Where does regularity stop? He proposed three classes: deterministic laws, holding without exception; strong probabilistic laws, where one correspondence occurs in more than half of cases; and weak probabilistic laws, where one correspondence is merely the most common, and could apply none of them, for want of counts.

He also named the mathematics that was already available: the matching problem of Montmort (1708), which asks how often two shuffled sequences agree by accident. The permutation nulls used throughout this paper are that problem solved by simulation rather than by formula. What is new here is not the method but the measurement: we compute the numbers on this material instead of asserting them, which is what Doerfer was asking for.

3 Resources

A learned model needs examples, and for this problem almost none existed as data. We built five datasets: three of transcription pairs, and two lexicons of Turkic itself. We release or make reproducible all of them; §12 sets out which is which and why.

3.1 Corpus 1: *Secret History of the Mongols*

The *Secret History* is a thirteenth-century Mongolian chronicle preserved only in Chinese characters used phonetically, with an interlinear gloss. It is the largest known parallel text of its kind, and it had not been extracted as aligned pairs. Parsing the digital edition yielded **9,329 Chinese–Mongolian transcription pairs**.

One detail decides whether the extraction worked. In the source encoding, a Chinese character followed by U+180B (a Mongolian free variation selector) marks an *annotation*, not part of the transcription. Treating those as data corrupted roughly a fifth of the corpus; the rule is one line, and finding it took considerably longer. Length agreement between the Chinese and Mongolian sides is 93.4%, which is the internal check that the alignment is real.

3.2 Corpus 2: Ligeti's Sino-Uyghur vocabulary

Ligeti[10,17] published a Ming-dynasty Chinese–Uyghur glossary in two instalments in *Acta Orientalia*.[17,18] The Turkic forms are given in Latin transcription and the Chinese in EFEO romanisation, which makes the text parseable. Extraction yielded **594 Turkic–Chinese pairs**.

We had to manage four properties of the typography: the apostrophe is U+2019 rather than ASCII, Turkic forms may contain spaces, footnote digits intervene before the gloss, and the Turkic form is not always at the start of a line. Our first pass missed them and returned

297 pairs, half of them wrong. A further trap was structural rather than typographic: journal running heads carry the author's name on verso pages, so a filter written for the article title alone silently discarded half the article.

3.3 Corpus 3: verified Buddhist Sanskrit transcriptions

Buddhist translators into Chinese used two devices: **音譯**, spelling a Sanskrit word by sound, and **意譯**, translating its meaning. Only the first is transcription. A 27,110-entry terminology dictionary[1] mixes them without marking which is which, and the two are not separable by inspection.

We had two approaches fail before one worked. Character-set membership admits **福慧** for *puṇya*, which is a calque. Exclusivity, treating a character as phonetic only if it never appears in a translation, collapsed the corpus to 67 pairs. What worked was phonetic verification of scoring each candidate against the Later Han reading of its characters and keeping only those whose sound matches. The result was **1,017 verified pairs**. We kept the 7,480 rejected entries separately so anyone can audit the decision. The induced **音譯** inventory matches the standard akṣara table, which is an external check providing that the filter found something real.

3.4 Lexicons 1 and 2: Kāşgarî and the *Codex Cumanicus*

The tests in §6 need dictionaries of Turkic itself, from near the period, against which a reading can be looked up. We found no downloadable dataset, so we extracted the following dictionaries.

Kāşgarlı Mahmud's *Dīwān Lughāt al-Turk* (1072–77) is the earliest dictionary of any Turkic language.[12] The Türk Dil Kurumu index volume has a machine-readable text layer; parsing it yielded **6,820 headwords** with Kāşgarî's Turkish glosses and his volume-and-page references.

The *Codex Cumanicus* (c. 1300) records Kipchak Turkic, compiled by Italian merchants and German missionaries, outsiders writing down a language they did not speak, which is the position the Chinese scribes were in. Géza Kuun's 1880 edition prints a Cumano-Latin vocabulary; parsing it yielded **2,110 headwords** with Latin glosses.[14] We deliberately skipped the Persian and German vocabularies that follow in the same volume. The Persian one would inject Iranian vocabulary into a Turkic lexicon and inflate every match rate computed from it.

4 Method

The channel we built is a probabilistic model of the Chinese scribe: for a source sound, what Chinese spelling would he choose? Character-to-sound alignments are learned by expectation–maximization over the training pairs, without hand-alignment. Emission tables are *reading-anchored* with backoff: full reading, then onset plus rime, then onset, then a global distribution, so a character never seen in training still receives a distribution rather than nothing.

Decoding runs the channel backwards: given the characters, score candidate source words. The binding variable throughout is not corpus size but **observations per unit**: how many times the model has seen each thing it must estimate. That figure, not the number of pairs, is what predicts whether reconstruction works.

We paired every comparison in this paper with a null. For lexicon searches we built pseudo-names by resampling Later Han syllables from the same names being tested, so they carry the right shape and the right sound inventory and no meaning at all; the identical search is run on them. For language comparisons we subsample the corpora to equal size before scoring, because otherwise the thinner model wins by being vague. Negative controls, meaning the method run on cases whose answer is already known, are what exposed the failures reported in §6.

5 What channel recovers

Every figure below is on held-out data. "Exact" means the model reconstructed the whole word correctly; "within 2" means it got no more than two segments wrong.

TRAINED ON → TESTED ON	EXACT	WITHIN 1	WITHIN 2
Mongolian → Mongolian (unseen chapters)	**36.9%**	66.3%	87.5%
Turkic → Turkic	**30.6%**	68.2%	83.5%
Sanskrit → Sanskrit	25.0%	57.9%	77.0%
Sanskrit → *Turkic* (wrong training language)	**3.1%**	**6.9%**	**27.5%**

TABLE 1 Each row is the result of one experiment. Row 1 trains on some chapters of the *Secret History* and tests on chapters held back entirely, a document-level split, so no word from a test chapter influenced training. Rows 2 and 3 do the same within the

Turkic and Sanskrit corpora, but those are glossaries without document structure, so the held-out portion is a random sample of pairs. Row 4 is the diagnostic: the channel is trained entirely on Sanskrit and asked to read Turkic. A model trained on Buddhist transcriptions barely functions on Turkic ones, 3.1% against 30.6%, which is the clearest evidence that the channel learns something specific to the language it was trained on rather than a generic Chinese-to-anything mapping.

CORPUS	PAIRS	OBSERVATIONS PER CHARACTER
Mongolian	9,329	~60
Sanskrit	1,017	~12
Turkic	594	~7
Xiongnu names	39 items	1.4

TABLE 2 The Xiongnu material is 112 character-tokens over 81 distinct characters. This is the number that governs everything in §6: the method does not fail on Xiongnu because Xiongnu is mysterious, but because 1.4 observations per character is not enough to estimate anything. The comparison is with 60 for the corpus where the same method reaches 36.9% exactly.

6 What the evidence cannot support

Two claims commonly made from this kind of material do not hold up and establishing that carefully is important.

6.1 Determining which language a name belongs to

The obvious application is settling whether the Xiongnu ruling house spoke Turkic, Yeniseian, Mongolic, or something else. We built the comparison properly: score how well each candidate language explains the transcription, equalise corpus sizes, compare by Bayes factor.

It failed its own controls. Given Ming Uyghur transcriptions whose language is known as Turkic with certainty, the method identified Turkic correctly in 69% of trials against a 50% chance baseline, and in 2 of 24 experimental cells it fell *below* chance. A diagnostic showed why: the phonotactic priors available for these families, taken from ASJP,[29] are close enough to each other that the comparison is measuring the reference data rather than the names. A procedure that cannot reliably identify a language it has been given cannot be trusted to identify one nobody has.

The strongest advocate of the rival hypothesis reaches the same conclusion on methodological grounds rather than measured ones. Vovin, arguing for a Yeniseian Xiongnu, sets the titles and personal names aside as evidence altogether: most of the glosses the Chinese sources preserve are of that kind, he calls them "practically useless" for determining genetic affiliation, and he declines to evaluate the ones Pulleyblank had used, on the ground that such evidence is not admissible.[27] The record this paper works from is almost entirely titles and personal names. Our result agrees with his judgement and replaces it with a number.

6.2 Reconstructing the Xiongnu names in full

At 1.4 observations per character the decoder returns candidates, but they are not constrained by evidence. An intermediate version of this work gated its output on in-domain accuracy and would have printed confident readings (**撐犁** as *ara*, *kara*, *tara*) where true out-of-domain reliability was 3.1%. Re-gating on out-of-domain accuracy removed them. This was the closest the project came to publishing a wrong answer, and we record the correction here because the failure mode is general: a decoder always returns its best candidate, and best is not the same as supported.

6.3 Matching readings to dictionary words

There is an obvious next step, and it is the one every reader wants to take. Once **稽粥** is written out as *kei çuk*, it is hard not to hear Turkish *küçük* "small, younger". The method is simple: look the reading up in a dictionary and see what it matches.

The trouble is that it always works. Modern Turkish has twenty thousand words in common use, and a target with three consonants has hundreds of near neighbors. The question is never "is there a match" but "is there more of a match than a string that cannot mean anything would get?" We built seven lexicons and gave each its own null.

The modern side of the comparison uses open spelling dictionaries rather than a published dictionary's headword list because a spelling dictionary can be counted. Turkish is the *hunspell-tr* word list, whose 292,000 forms are cut to 21,721 by keeping only words with a wordfreq Zipf score of at least 2.5, that is, words a speaker actually uses.[25,23] Kazakh is the *myspell-kk* list, 53,182 forms, romanised on load by a fixed Cyrillic-to-Latin table.[13] The frequency filter matters: without it the Turkish list is dominated by inflected and rare forms, and every target finds a match. The historical side is described in §3.4.

LEXICON	ENTRIES	PSEUDO-NAMES FINDING A MATCH	MEDIAN SCORE FOR A PSEUDO-NAME	NAMES BEATING CHANCE, P ≤ 0.05
Turkish, common use	21,721	231 / 250	0.156	2 of 36
Kazakh	53,182	239 / 250	0.183	0 of 36
Union of Turkish and Kazakh	73,406	236 / 250	0.117	2 of 36
Turkish, loan-shaped words removed	14,876	233 / 250	0.200	2 of 36
Intersection (in both languages)	1,497	219 / 250	0.329	3 of 35
Dīwān Lughāt al-Turk, 1072–77	6,337	276 / 300	0.200	3 of 36
Codex Cumanicus, c. 1300	2,067	275 / 300	0.242	3 of 36

TABLE 3 A lower score is a closer match, so a *lower* median in column four means the lexicon is noisier: meaningless strings are finding better matches in it. Read that column as the price of admission. Two results are worth separating out. **Pooling two dictionaries makes matters worse**: the union finds more matches for the real names and more for the meaningless ones, and the noise floor falls from 0.156 to 0.117. A larger haystack does not help you find a needle. What helps is the opposite operation of keeping only words present in *both* languages. This lifts the floor to 0.329, because a word must survive two independent borrowing histories to qualify. Unfortunately, the intersection is over identical spellings, and cognates diverge, so Turkish *tanrı* and Kazakh *täñir* both fall out while *domino* and *kauçuk*, borrowed into both recently, survive. This enriches for shared modern loan words. Across all seven lexicons, between none and three names in thirty-six could meet p ≤ 0.05 significance, which at a one-in-twenty threshold is what thirty-six coin flips give.

The older lexicons do better than modern ones on two cases with a known answer.

NAME	READING	LEXICON	RANK	CANDIDATE	GLOSS IN THAT DICTIONARY
撐犁 chengli	ḍaŋ lei†	Modern Turkish	–	not returned	*dengeli* "balanced" came first
		Dīwān Lughāt al-Turk	4th	tengri	*gök, sema* ("sky, heaven")
		Codex Cumanicus	2nd	tengri	*deus* ("God")
頭曼 Touman	do mɑnᶜ†	Modern Turkish	–	duman	"smoke"; *tümen* not in the top three
		Dīwān Lughāt al-Turk	3rd	tümen	*pek çok* ("very many")
		Codex Cumanicus	2nd	touman	*nebula* ("mist"); *tümen* absent from this lexicon

TABLE 4 The *Hanshu* glosses 撐犁 as 天 "heaven". Kâşgarî in Baghdad and the compilers of the Codex in the Crimea, working a thousand years later with no knowledge of the Chinese record, gloss *tengri* as "sky, heaven" and as *deus*. The attestations are independent and the search connects them, which fails to do against a modern dictionary, and does better the older and smaller the lexicon becomes. The second name repays attention because it cuts against the accepted etymology: *tümen* "ten thousand" is absent from the Cuman vocabulary entirely, and in every lexicon holding both, the search prefers a different lexeme: Turkish *duman*, Kâşgarî's *tuman*, Cuman *touman*, all "smoke, fog, mist", one word across three lexicons and a thousand year separation. The difference is one vowel, and §7 puts vowel recovery at 50%. This is recorded because the alternative has never been raised.

A caution about the strongest datum. The match between `ḍaŋ lei` and *tengri* is the most quoted evidence for a Turkic Xiongnu. The inference only works if *tengri* is Turkic to begin with. The word is not confined to Turkic: Mongolic has *tenger* "sky", Turkic *tengri* survives as modern Turkish *Tanrı*, and Georg has argued that the Turko-Mongolic word is built on Proto-Yeniseian `*təŋgVr-` "high" with a Turkic suffix, making it a borrowing rather than an inheritance.[8,28] Georg's derivation is one argument among several and need not be accepted; what holds either way is that a word attested across Turkic, Mongolic and, on Georg's account, Yeniseian cannot discriminate between them. **撐犁** shows that the Xiongnu used, for "heaven", a word also found in Turkic and Mongolic. It does not show which of those languages they spoke.

7 Answering the 1974 question

The Xiongnu names survive in Han-period sources; the Turkic training corpus we used is Ming era, which is twelve hundred years later. We can measure directly whether anything carries across that gap. We took characters whose readings are known in both layers, predict the earlier reading from the later one, and score by component.

PART OF THE SYLLABLE	RECOVERED CORRECTLY MING → LATER HAN	95% CONFIDENCE INTERVAL	CHANCE BASELINE: BEST BLIND GUESS	LIFT: POINTS GAINED OVER THAT GUESS	CLASS OF SOUND LAW[15]
Nasal ending (-m, -n, -ng)	99.2%	95.4 – 99.9	35.4%	+63.8	deterministic
Initial consonant	79.3%	72.4 – 84.8	9.1%	+70.1	strong probabilistic
Vowel	50.0%	42.4 – 57.6	15.2%	+34.8	weak probabilistic
Stop ending (-p, -t, -k)	0.0%	0.0 – 7.9	35.4%	−35.4	no law

TABLE 5 n = 164 characters (nasal ending 118, stop ending 45). "Recovered correctly" is leave-one-out majority prediction of the earlier reading from the later; intervals are Wilson score intervals; the chance baseline is the unconditional majority class for that part, the score obtained by always guessing the commonest value. **Lift is column two minus column four**: how many percentage points knowing the later reading provides. The aggregate coda figure shouldn't be used. Blending nasals and stops gives a flattering 87%, which means nothing, because the sample is 72% nasal-ending characters.

This is the reviewer's question, answered. Three of the four components classify cleanly. Nasal endings, at 117 of 118, are deterministic for practical purposes. Initial consonants, at 79.3% with the interval nowhere near one half, are a strong probabilistic law. Stop endings are not a law in any sense: not one of 45 survived, and the interval tops out at 7.9%.

The fourth exposes a flaw in the rule itself. Vowels land at **50.0%** (82 of 164, on the boundary to the decimal), and the confidence interval, 42.4 to 57.6, straddles it. By the reviewer's criterion the vowel cannot be classified, and no larger sample would help, because the estimate is certain; it is genuinely at one half.

But one half is the right threshold only when there are two outcomes. A vowel is a choice among many vowels, and the honest comparison is against what you would get knowing nothing, the majority class, here 15.2%. Against that, vowels carry **+34.8 points** of real information: While it is a weak probabilistic law, it is not a coin flip. Stop endings, by the same measure, fall *below* their baseline. Knowing the Ming reading makes you worse off than guessing, because Old Mandarin merged -p, -t and -k into nothing and the merger is not recoverable.[24]

So the fifty-year-old question has a shape that the fifty-year-old debate could not see. The threshold is not a constant. It is the baseline of the component in question, and it moves with how many outcomes that component has.

8 Three measurements of scribal practice

8.1 Is a Chinese `m-` ever a foreign `b-`?

Reading 冒頓 as *Bayatur*, the most repeated etymology in the field, requires the `m-` to stand in for a `b-`. Proto-Turkic has no initial `*m-`: the `m-` words in Old Turkic are loans or come from `*b-` assimilating to a nasal immediately after the vowel: **bän* → *men*, **buŋ* → *muŋ*.[4,7] So on the Turkic reading the substitution is load-bearing. Each corpus is a set of transcriptions whose source word is known, which makes the question a counting exercise.

CORPUS	SOURCE WORDS BEGINNING `B-`	WRITTEN WITH A CHINESE `M-` CHARACTER
Mongolian (*Secret History*)	738	0
Turkic (Ligeti glossaries)	78	0
Sanskrit (Later Han, verified)	22	3 (13.6%)
Total	**838**	**3** (0.4%)

TABLE 6 The reverse direction is clean: Chinese `m-` characters render a source `m-` in 303 of 305 Mongolian cases, 90 of 97 Sanskrit and 14 of 17 Turkic. The forward direction, which is the one that matters here, must be read by layer and not pooled. The total of 3 in 838 is 0.4%, but all three instances fall in the *Later Han* corpus, where the

rate is **3 of 22, or 13.6%**, with a Wilson interval of 4.7 to 33.3. The two later corpora contributed 0 of 816 between them. Pooling across eight centuries of Chinese produces a figure that describes no period at all, which is the point §7 makes in general terms.

Read by layer, the substitution is not exceptional. A rate of 13.6% is the same order as `ts-` for a source affricate, which §6.3 treats as a minority but attested spelling at 15%. The scribe did have the alternative: within the Later Han corpus a source `b-` took a `b-` character 26 times in 32. But having an alternative and always using it are different things, and once in seven he did not. **So, an `m-` character is not, by itself, an argument against a Turkic reading of 冒頓.**

The onomastic record points the same way, in a later period. Tang scribes wrote the Türk ruler known in the inscriptions as Qapaghan Qaghan with the personal name **默啜**, read as *Bögü-Çor*; and the Uyghur ruler **牟羽** is the same word *bögü* "wise" behind a different `m-` character. Two different Chinese `m-` characters for one Turkic `b-` word is a pattern, not an accident. A third Türk name often cited alongside these two, **木杆** for Muqan Qaghan, is a control rather than a third instance: the Sogdian Bugut inscription writes that name *mwγʾn*, also with `m-`, in a script that has a letter for *b*, so the name's own initial is disputed and cannot be assumed to be `b-`. These are Tang rather than Han, and Late Middle Chinese treated its nasal initials differently, so they corroborate that the practice existed rather than fixing its Han-period rate; but they remove any sense that the substitution would be unheard of in a ruler's name.

The objection moves to the other end of the word. *Bayatur* ends in `-r`, and **頓** `tuənᶜ` carries a written `-n`. Later Han had no `-r` coda, so a scribe facing a source syllable such as *tur* had to choose something, and the Sanskrit corpus records what he chose. Of 37 source syllables ending in `-r`, he left the syllable **open** 21 times (57%), used a `-t` coda **14** times (38%): **達** for *-har* in *bodhidharma*, **羯** for *kar-* in *ākarṣaṇī*. A `-p` coda occurs once, and a `-n` coda **once**, 3%. Reading **頓** as *tur* therefore asks for the rarest of the four options, where the expected spelling was an open character or a `-t`.[22] The syllable count is no obstacle to either reading. *Bayatur* has three syllables against two characters, but 93 of the 1,017 verified pairs use fewer characters than syllables, and in 47 of those a character with a written stop coda does the compressing. **竭** covers *-gata* in **多阿竭** for *tathāgata*, **塞** covers *-saka* in **優婆塞** for *upāsaka*. Compression is a real device, available to any reading of **冒頓**, so the final `-r` stands alone as the difficulty.

What survives on the Turkic side is unaffected by any of this. Proto-Turkic has no initial `*m-`, and the one route to one is `*b-` assimilating to a nasal immediately after the vowel; **冒** `mək` has no nasal in that position, its only nasal being the `-n` of **頓**. So, on a Turkic reading the `m-` must be the scribe's choice rather than the source language's sound, and the count above (13.6%) says he was entitled to make it. Small numbers should be kept in view: 3 of 22 is four cases short of nothing, and the interval is wide.

8.2 How was a source coda spelled?

A related question, and the one §9 turns on. When a source syllable ended in a consonant, did the scribe use a character ending in that consonant, or an open character with a vowel after it? The Sanskrit corpus answers it for the right period: for the 770 pairs aligning one character to one syllable, each Sanskrit syllable is matched to its character and the coda of each recorded.

SOURCE SYLLABLE ENDS	N	CLOSED WITH THE SAME CODA	WRITTEN WITH AN OPEN CHARACTER
-n	93	**83%**	13%
-ŋ	22	**73%**	9%
-t	53	53%	28%
-k	21	48%	38%
-p	10	20%	20%
All stops	**84**	**48%**	**30%**

TABLE 7 Roughly a third of stop-final source syllables were written with an open character, against 13% and 9% for nasals. An open spelling after a stop is ordinary, not exceptional. Note the shape of the asymmetry: nasals are handled robustly and stop loosely, the same split Table 5 found surviving the period gap, appearing here in what the scribe chose to write rather than in what survived. Two independent measurements of one underlying fact.

8.3 When a consonant is written twice, was it said twice?

A scribe could spell one source consonant across a syllable boundary: the first character ends in `-t`, the second begins with `t-`. Whether doubling records a geminate or is a convention of the script changes how many consonants a transcription claims, and therefore how a name may be read. It is answerable on the Sanskrit corpus, where 772 pairs align one character to one syllable.

We take every junction at which a written stop or nasal coda is followed by a same-place onset, and ask whether the Sanskrit form carries that consonant once or twice:

JUNCTIONS WHERE A WRITTEN CODA IS FOLLOWED BY AN ONSET OF THE SAME PLACE	THE SOURCE WORD HAS THAT CONSONANT ONLY ONCE	THE SOURCE WORD HAS IT TWICE
140	**121 (86%)**	19 (14%)

TABLE 8 The double spelling is a convention, not a geminate: 釋迦 for *śākya*, 怛他 for *tathā*, 富單 for *pūtana*. A sequence such as 骨都 `kuət tɑ` therefore transcribes a source *kuta* with one `t`, not *kutta*. The count is conservative: "twice" is scored whenever the doubled letter appears anywhere in the Sanskrit form, so a geminate elsewhere in the word counts against the merger, and 86% is a floor rather than a ceiling. Like §8.2, this is a measurement that settles an objection which would otherwise be argued from intuition: a reader who assumes two written consonants mean two spoken ones will reject readings that the corpus permits. The nasal case is cleaner still: every source word in the corpus with a medial `ŋg` or `ŋk` is written with a `-ŋ` coda followed by a velar onset, 13 of 13: 央具梨 for *aṅguli*, 商羯羅 for *śaṅkara*, 僧伽施 for *sāṅkāśya*. Two characters spelling one medial nasal is not a license the scribe occasionally took; it is the only device available to him, since a Chinese syllable cannot begin with `ŋ`.

9 A worked example: what a reading proposal must survive

Section 6.3 shows that reading a transcription against a dictionary is, on its own, at chance. That is not an argument that no such reading can ever be supported; it is an argument that support must come from somewhere other than the resemblance. One case in this material meets that standard, and it is set out here as a worked example rather than as a result about Xiongnu.

From 呼韓邪 onward, every chanyu title ends in 若鞮, Later Han `ńɑk te`. *Hou Hanshu* states that the element was added to the title from that reign and glosses it 孝 "filial". This is the only Xiongnu title element the Chinese sources translate, which means the meaning is *recorded*, not inferred from the sound.

Four things then must hold, and each is checkable independently of the others.

- **The sense must be attested.** We searched the two lexicons of §3.4 and found *Inak*, glossed *fidelis* "faithful", in the *Codex Cumanicus*.[14] It is absent from Kāşgarî, so the word belongs to the Kipchak tradition. Old Turkic *ınaq* "trusted one, confidant" is generally derived, with *ınan-* "to trust", from a root *ına-* "to rely on". The sense the proposal needs is documented in a medieval glossary, and it matches 孝 independently.
- **The word is of the kind that becomes a title.** This matters more than the gloss alone. Clauson records Chagatay *inak* as *nāyib va muqarrab*, "a royal representative or senior minister", and the cognate *ınanç*, from the same *ına-* root, appears in Old Turkic as an office and as a name element: *él inanç tirek* in a Uyghur list of proper names, *ınançları buyrukları* "his ministers and officers", *inanç tayanç* for a confidential minister to a *beg*, and *él ögesi ınancu bilge* in the Yenisei epitaphs of southern Siberia.[4] A court title built on this root is therefore not a stretch imposed on the word; it is what the word ordinarily does. *Hou Hanshu* has 若鞮 entering the formal name from a particular reign onward, which is exactly how such a title spreads.
- **The morphology must be licensed.** A reading *ınaq-tı* is not available: *-tI* is not a noun-forming suffix, and the lexicons show it rather than a grammar having to assert it. Of the 31 Cuman headwords ending in *-tI*, almost all are glossed with Latin perfects (*Ayti: dixit*, *Azti: transgressus est*, *Öwtiti: detersit*), the definite past ending; the remainder are numerals or Arabic loans. Kāşgarî's index has six, none containing a suffix. A title cannot be a past tense verb. Old Turkic does have a suffix that fits: *-t*, a plural used specifically with *titles*: *tegin* → *tegit*, *tarqan* → *tarqat*.[7] On that reading the element is **Inakt**, "the faithful ones", and 若鞮 is not one ruler's epithet but a title borne by a line, which is exactly how the Chinese sources use it.
- **Objections to the spelling must be tested, not asserted.** Against a final `-t` it can be objected that Later Han had `-t` final syllables and a scribe would have used one rather than the open 鞮 `te`. That is a claim about scribal habit, and

Table 7 counts it: 30% of stop-final source syllables were written with an open character. Hence the objection does not hold.

- **What remains unexplained must be stated.** If the source was *ınakt*, the opening *ı*- is not written, and 若鞮 has no room for it. That objection stands.

Three of four hold, with the fourth named. That is the standard this paper asks of a reading proposal, and it is deliberately higher than resemblance. The sense is attested in a source contemporary with neither party to the comparison, the morphology is licensed by a suffix with the right distribution, and the one objection that could be tested was tested and failed.

One further reading of our own, 冒頓 as *Bögü Tın*, meets this standard on one of the two Later Han readings of 頓, and §9.2 sets out what that turns on. We produced 31 others in the course of this work. None of them meet it (by §6.3 they are at the chance rate), and they are therefore **not presented here as findings**. We include them as **Supplementary Material S1** and release them in the repository as `author_proposals.csv`, dated and labeled, because readers will run this search for themselves regardless and it is better that the output is written down than imagined.[26]

We supply S1 for the record, not for review. It has the status of a laboratory notebook that happens to be legible. Each row gives the Chinese form, the Later Han reading, the author's rendering, the sense proposed for it, and, where one exists, the historical or onomastic parallel that suggested it. Those parallels are drawn mostly from later Turkic naming practice, where a ruler's epithet is a common noun (Bayezid *Yıldırım* "the thunderbolt", Alp Arslan "the lion", Mehmed *Avcı* "the hunter"), and they are offered as the kind of thing that would have to be argued rather than as an argument. A reader who accepts none of them loses nothing from the paper. The measurements in §§5–8 do not rest on any of them. The one reading that is argued here, *Inakt*, is in S1 too, marked as the exception, so that the file can be read as a single list with the standard visible in it. S1 also contains a worked demonstration, at §S1.5, of why the decoder's own output must not be quoted in support of an individual reading, the concrete form of the warning in §6.2.

9.1 Two cases which are not ours to propose

We can apply the same standard to comparisons somebody else has already made. The selection bias §6.3 measures is removed by construction if the reading was proposed by others. Two items in the record carry a reading from the literature and a Chinese gloss to check it against.

The first is 徑路 `keŋᶜ laᶜ`, the 徑路刀, a Xiongnu blade used in oath-taking. Friedrich Hirth compared it with Turkic *kıŋrak* in 1908, working from the *Hanshu* account of the treaty of 47 BCE, where the scholiast glosses the word as the precious sword of the Xiongnu. He drew the Turkic side from living Turki dialects rather than from Kāşgarî, and extended the identification to 輕呂, the word the *Zhoushu* uses for King Wu's dagger, which is why he called it the oldest Turkic word on record. That further claim is his and is not tested here.[11] Kāşgarî glosses *kıñrak* as a large knife resembling a cleaver, for cutting meat and dough.[12] The Chinese side independently says the referent is a blade, two sources with no contact agreeing about what the object is, which is a stronger kind of agreement than a sound match. The consonant frame is exact: 徑 gives the velar onset with its `-ŋ` coda written, and 路 gives a Chinese `l-` for a source `r-`, the direction attested in 12 of 14 cases. The Turkic word's final `-k` is not written, and Table 7 gives the rate at which a stop-final source syllable was spelled with an open character: 30%.

Where the Turkic word comes from. Clauson derives *kıŋra:k* as a deverbal noun from an unattested **kıŋra:-* (the colon is Clauson's mark for a long vowel), itself denominal from *kıŋır*, in the sense of something curved or something that cuts crookedly.[4] Kāşgarî's own index supports that family on the same frame: *kıñır*, *kıñru* "askew, squint", *kañrak* "palate", the arch of the mouth, and the verb *koñur-* "to wrench, bend back", surviving as Turkish *kanırmak*; and the *Codex Cumanicus* independently glosses *Kingir* as *curvus*. A curved blade named from bending is therefore motivated inside Turkic by the standard authority, not by us.

The word also survives. Clauson records it in Teleut as *kıŋırak*, and in Kirghiz as *kıŋarak / kıŋırak*, "a rough two-edged knife used for cutting felt, scraping hides and sheepskins", the same object, in living languages, a thousand years after Kāşgarî and two thousand after the *Hanshu*.

Kāşgarî's text at volume III, page 382 of the *Dīwān* reads *kırpa:k* and is corrected in the margin to *kıŋra:k*. So this form rests on a marginal emendation rather than the text as written. The emendation is not the only support for it: Hirth took the word from living Turki dialects in 1908, and the Teleut and Kirghiz survivals cited above are independent of the Dīwān's manuscript altogether. *kıñır*, on the other hand, fits the transcription exactly, at three consonants with nothing supplied; *kıñrak*, which means a knife, needs the unwritten final `-k` that Table 7 licenses at 30%. The form that fits best does not mean "blade", and the form that means "blade" does not fit best.

> **A rival account.** The same object has been derived instead from an Iranian word for a short sword, the type whose Greek name is *akinakes*. The history supports it: the Xiongnu's western neighbors were Iranian-speaking, and the weapon is present in steppe archaeology, so a borrowed name is not a stretch.
>
> Unusually, that account is testable with the same machinery, and it does not survive as well. 路 `la`^{C} is a liquid-onset character, so the channel requires a liquid in the second position of the source word. The *akinakes* family (`k–n–k`) has no liquid anywhere in it. If the proposal is that 徑路 transcribes that word, it fails on the frame, not at the margins.
>
> We have not seen the exact Iranian etymon its proponents advance, so the test applies only to the form named. Our channel establishes what a transcription *permits*, never what it requires: a Turkic reading surviving does not make a Turkic source the only one available. This is Doerfer's objection in its sharpest form. The honest report is that the Turkic candidate survives its test and the Iranian one, as stated, does not.

The second is 屠耆 `dɑ gɨ`, which the Chinese sources gloss 賢 "worthy, wise" and which appears in **左右屠耆王**, the Wise Kings of the Left and Right. The Turkic-hypothesis literature reads it **toγrï*. 耆 is one of the daggered characters: the table gives `gɨ`, `ki`^{B} and `tśi`^{B}, and the comparison below uses the first. The second would serve it equally well; the third would not. The *Codex Cumanicus* has *Togru* glossed as *rectus*, and the skeleton matches at the same single step, a Chinese `l-` for a source `r-`.

The semantic fit is closer than "straight" alone suggests. Clauson derives *toğuru: / toğru:* as a gerund of *toğur-*, physically "straight" and then metaphorically "straight, honest, upright, true", and that moral sense is old rather than modern. A fourteenth-century glossary renders it with Arabic *al-tiqa* "trustworthy, honest", and *toğru ayt-* with *ṣadaqa* "to tell the truth".[4] The word survives across the whole family and as modern Turkish *doğru*. So, the comparison is not between 賢 "worthy" and a word for physical straightness, but between 賢 and a word whose ordinary sense is "upright, honest, trustworthy".

The root family also yields names, which is the argument §9 makes for *inak*: Kāşgarî records that men are called *Toğrıl* after a bird of prey, an *Alp Toğrul Tégin* appears in a Uyghur list of proper names, and the first Seljuk sultan bore the name. What remains soft is the vowel, and the fact that 144 attested headwords fit this frame, so the transcription does little to narrow the choice.

So, two existing comparisons survive tests that 32 of our own readings fail, and one of the two has a rival etymology our method cannot rule out. That asymmetry is the point of this section. The channel cannot generate supported readings at this data density, but it can judge ones already on the table, and deciding on other people's proposals is a use of it that the accuracy figures actually license.

9.2 Where a century-old disagreement actually lies: 冒頓

The counts of §8.1 constrain a reading of 冒頓 from both ends. 冒頓 is the most consequential name in the record, and the frame it fixes looks unusually tight, because **both codas are written**. The second of them, however, is not a single value. Schuessler's table gives 頓 two Later Han readings, `tuən`^{C} and `tuət`, one ending in a nasal and one in a stop, and the reconstruction column of Appendix A prints the first. That choice, which no published reading of this name states, is what separates our proposal from the field's.

The first character has a documented Turkic value. This is not an inference from a rate, and it holds whichever reading of 頓 is taken. In the Later Han table 冒 and 默 have the *same* reading, `mək`. And 默啜, the personal name of the Türk ruler known in the inscriptions as Qapaghan Qaghan, is read *Bögü-Çor*. So that exact Later Han syllable is on record writing *bögü* in a Turkic ruler's name: the `m-` standing for the source `b-`, the written `-k` standing for the source `g`, and the final vowel of *bögü* not written at all. The same substitution is on record more than once in Turkic onomastics, and the characters used are worth setting out: 默 `mək` in 默啜 for *Bögü-Çor*, and 牟 `mu` in 牟羽 for the Uyghur Bögü Qaghan. Both write *bögü* with a Chinese `m-` character. 冒 is itself listed with a second reading, `mou`^{C}, and the two attested spellings of *bögü* fall either side of that split: 默 matches 冒's first reading exactly, and 牟 `mu` is the nearer of the two to its second. Whichever row of 冒 is taken, there is a Turkic precedent for that character writing *bögü*.

Each step of that is separately licensed by the corpora. The `m-` for a source `b-` is 3 of 22 in the Later Han layer, 13.6% (§8.1). One character covering two source syllables, with a written stop coda doing the compressing, occurs only in 47 of the 1,017 verified pairs: 竭 for *-gata* in *tathāgata*, 薩 for *-sattva* in *mahāsattva*, 弗 for *-putra* in *śāriputra*. 冒 for *bögü* is that same shape.

The second character decides the name, and it has two readings. On `tuən`^{C} the frame ends in a nasal. The two lexicons then supply *tın* "spirit, breath" (Codex *Tin*, *anima*), *tün* "night" (Codex *Tun*, *nox*), *tan* "dawn" and *tun* "rest", every one of them matching with nothing inserted and nothing discarded, and the name reads ***Bögü Tın***, "sage spirit". On that branch *Bayatur*, the field's answer for a century, fails on the one point such a frame decides cleanly: it ends in `-r` against a written `-n`, a spelling that occurs once in 37 opportunities.

On `tuət` the frame ends in a stop, and both verdicts reverse. A source syllable in `-r` is written with a `-t` coda in 14 of those same 37 opportunities, 38%, so *Bayatur* is not merely permitted but ordinary. And our own second element becomes *tut* rather than *tın*, which the lexicons do not support in the same way.

> **The disagreement is smaller than it looks, and it is not phonological.** Two readings of the founder's name have stood against each other for a century. They do not rest on different views of Later Han phonology, of what a Chinese scribe would do, or of Turkic word formation. They rest on which of two rows of Schuessler's table is taken for one character, and the transcription evidence cannot choose between them, because both rows describe the same character on the same page. What the channel contributes here is not a reading but the location of the disagreement.
>
> The conventional pronunciation of the compound, *Mòdú*, points to the stop rather than the nasal, which favours *Bayatur*, and we record that rather than passing over it. It is a reading tradition rather than a measurement, and it is not ours to adjudicate.

Measured against the standard of §9, and on the `tuən`^{C} branch only: the sense of both elements of *Bögü Tın* is attested independently of any Chinese source; the morphology is the commonest Turkic name shape, two nouns in apposition, as in Alp Arslan or *Bögü-Çor* itself; and the objection that would have killed it, a Chinese `m-` for a foreign `b-`, was measured and dissolved rather than asserted away. What remains unexplained is stated: the compound *bögü tın* is not itself attested, only its two halves; the vowel of the second element is not recoverable, so *tün*, *tan* and *tun* are equally available and choosing *tın* is a semantic preference rather than a phonological result; and the tone category of 頓 is not used. We prefer *tın* among the four because of what *bögü* is: Kāşgarî glosses it *bilgin, akıllı, hakim*. Clauson says the word "seems to connote both wisdom and mysterious spiritual power", and it was borrowed early into Mongolian as *böge*, the ordinary word for a shaman. It appears in regnal styles across the record: *Teŋri Bögü Teŋriken* in the Old Turkic inscriptions, Bögü Qaghan of the Uyghurs (**牟羽** in the Chinese sources), and Bögü-Çor of the Türks (**默啜**).

The choice is not peculiar to this name. Across the record, **46 of 107 characters carry more than one Later Han reading, and in all 46 the value printed in Appendix A is the first row of the table** (§11). **冒頓** is the case where that choice carries the most weight. The claim this section makes is therefore bounded: *Bögü Tın* survives every test we can apply on one of the two readings of **頓**, the field's reading survives on the other, and the transcription evidence does not choose between them. The material for choosing, the counts, the corpora, and the scripts, is in the repository that we provided.

10 Where the method does work

The constraint identified in §5 is data density, not the difficulty of the language. That suggests where the future work need to go next. Göktürk-era material has what the Xiongnu material lacks: a source language that is not in dispute, an independent script recording it, and enough of it to train on.

CHINESE, AS WRITTEN IN THE TANG HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	INDEPENDENTLY IDENTIFIED AS, IN ESTABLISHED SCHOLARSHIP
葉護	`jap ɣua`^{C}	`yap ğua`	*yabghu*, a princely title
可汗	`kʰɑi`^{B} `gɑn`^{C}	`kai gan`	*qaγan*
特勤	`dək gɨən`	`dık gıın`	*tigin*, prince
突厥	`tʰuət kyɑt`	`tuıt kyat`	*Türk*
阿史那	`ʔɑ ṣiə`^{B} `nɑ`	`'a şiı na`	*Ashina*, the ruling clan
黠戛斯	`gɛt kɛt sie`	`get ket sie`	*Qïrqïz* (Kyrgyz)

TABLE 9 **Neither middle column is model output.** Column two is Schuessler's published Later Han reconstruction applied to Tang-era words, deliberately the wrong period, which is the point of the exercise. The last column is not an output of this work either: each identification is long settled on evidence independent of the Chinese transcriptions, the Orkhon inscriptions giving *qaγan*, *tigin*, *yabghu* and *Türk* directly in Old Turkic script, several corroborated again in Sogdian, Arabic and Byzantine sources.[7,9] They function here as ground truth. That 葉護 → *yap ğua* is readable against *yabghu* despite a period mismatch of four centuries measures the conservatism of Chinese scribal practice, not linguistic continuity between the Xiongnu and the Göktürks, on which this material bears not at all.

Chinese records name a great many steppe peoples, and they fall into three groups for these purposes. **Tractable**: those recorded in the Tang histories whose language is known: **鐵勒** Tiele, **回紇** Uyghur, **葛邏祿** Qarluq, **突騎施** Türgesh, **黠戛斯** Kyrgyz, alongside the Göktürk material. These have transcriptions, a securely identified source language and published readings for the right layer; they are training data. **Readable but uninformative**: groups such as **丁零** Dingling and the Avars, whose Chinese readings can be recovered but whose language is unknown, so a reading cannot be checked. **Out of reach**: the Xiongnu themselves, and the Pechenegs, who are barely in the Chinese record at all. The distinction is not how interesting the group is but whether there is anything to validate against.

11 Limitations of this work

- **No independent philological review.** The computational method and its evaluation are ours to defend. Philology is not our work, and we have not had it checked for this paper. The Later Han readings are Schuessler's, the etymologies discussed are from the published literature, and the identifications in Table 9 are the established consensus. A specialist reading would strengthen the paper and might revise judgments in §6 and §9.
- **Corpus validation is internal.** The extraction rules are documented and internally consistent (93.4% length agreement for Mongolian, an induced akṣara inventory matching the standard one for Sanskrit), but no external reviewer has checked a sample against the sources.
- **The Turkic corpus is Ming-era**, roughly 1,200 years later than the Xiongnu material. §7 measures what those costs rather than assuming it away.
- **Reconstructions are inferences.** Schuessler, Pulleyblank and Níng are three scholars with three notations, and Baxter and Sagart a fourth.[2,19,20] So some of the apparent disagreement between them is notational rather than substantive. A single table is also not single-valued: **46 of the 107 characters in the record carry more than one Later Han reading**, and Appendix A prints the first tabulated row in all 46. Where the alternative would change a reading it is named in the entry, and §9.2 is the case where it changes most.
- **Names travel.** Even a perfect reconstruction would not settle ethnicity.[16,21] Turkic tribes bore Iranian names and Mongol rulers used Turkic titles; the caution in §6.3 about *tengri* is a specific case of this general problem.
- **Reproducibility has a software dependency.** The headword count for the Kāşgarî lexicon differs between machines, 6,820 against 6,880, according to the version of the PDF text-extraction library. It shifts the noise floor by 0.008 and changes no rank or conclusion.

12 Data and code availability

We package everything described here as one repository, so that anyone can regenerate every number and table in this paper from the raw inputs by running the numbered scripts in order. The repository is the durable output of this work and this paper describes it.[26]

```
chinese-transcription-channel/
├── README.md            what each output is, and how to regenerate it
├── LICENSE              code MIT
├── DATA_LICENSE.md      the three-way split described below
├── data/derived/        the corpora, the lexicons, and every measurement
├── src/                 steps 1–23 plus lookup_lexicons.py, each rerunnable alone
├── reports/             one summary per step, with the numbers quoted above
└── docs/                design notes, including approaches that were rejected
```

Supplementary Material S1. *Candidate readings of the Xiongnu record* accompanies this submission and carries the same content as `data/derived/author_proposals.csv` in the repository. We supply it for the record under §9 and do not offer it as a result of this paper.

Released, CC BY 4.0. The Mongolian and Sanskrit corpora, the Cuman lexicon (Kuun's 1880 edition being public domain), the induced akṣara table, the Xiongnu record of Appendix A, and every measurement file behind Tables 1–9.

Withheld, with the extraction script released instead. Four derived files inherit restrictions from their sources: the Turkic pairs, from Ligeti's glossaries (© Akadémiai Kiadó); the Kāşgarî lexicon, from a modern Türk Dil Kurumu edition; and two files built on that lexicon: the Old Turkic echo results and the candidate-space listing, the latter also quoting Clauson's *Etymological Dictionary* verbatim in a context column. Anyone holding the sources can regenerate any of them byte-for-byte by running the numbered script, which keeps the paper's numbers checkable without redistributing the source. No permission is required for this arrangement, and none has been sought.

Not in the repository. Third-party source texts and datasets (dictionaries, scanned volumes, Unihan, the DILA authority files) are inputs, not outputs. Where a script needs one, its docstring says which file and where to obtain it.

A Appendix: The Xiongnu record

All thirty-nine names, titles, and lexical items in the Chinese histories, with the Later Han reading of each character and a Turkish-orthography rendering for readers who do not read phonetic notation. The Later Han column is published scholarship, not model output; our contribution is the assembly: we collected the names in one place and joined the readings to them character by character. Superscript letters mark Han tone categories. A dagger, †, marks a character for which Schuessler's table gives more than one Later Han reading; the first tabulated row is the one printed. **46 of the 107 characters in the record are so marked**, and where the alternative would change a proposed reading it is named in the Supplementary Material rather than left to the dagger. 若鞮 is listed once, shaded, rather than repeated in the six titles that carry it.

A.1 Rulers and the ruling clan

NAME OR CLAN NAME, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	DATES OF RULE, AND THE SOURCE IN WHICH THE FORM APPEARS
頭曼 Touman	do mɑn^{C†}	do man	209 BCE
冒頓 Modu / Modun	mək† tuən^{C†}	mık tuın	209–174 BCE
稽粥 Jizhou / Jiyu	kʰei^{B†} tśuk	kei çuk	174–160 BCE
軍臣 Junchen	kun dźin†	kun cin	160–126 BCE
伊稚斜 Yizhixie	ʔi ḍi^{C} ja	’i di ya	126–114 BCE
烏維 Wuwei	ʔɑ wi	’a vi	114–105 BCE
詹師廬 Zhanshilu	tśam ṣi liɑ	çam şi lia	105–102 BCE
呴犁湖 Xulihu / Goulihu	huo† lei† gɑ	huo lei ga	102–101 BCE
且鞮侯 Qiedihou	tsɑ^{B†} te† go	tsa te go	101–96 BCE
狐鹿姑 Hulugu	ɣuɑ lok kɑ	ğua lok ka	96–85 BCE

NAME OR CLAN NAME, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	DATES OF RULE, AND THE SOURCE IN WHICH THE FORM APPEARS
壺衍鞮 Huyandi	gɑ jan^{B†} te†	ga yan te	85–68 BCE
虛閭權渠 Xulüquanqu	kʰɨɑ liɑ gyan gɨɑ	kıa lia gyan gıa	68–60 BCE
握衍朐鞮 Woyanqudi	ʔɔk jan^{B†} guo te†	’ok yan guo te	60–58 BCE
呼韓邪 Huhanye	hɑ† gɑn ja†	ha gan ya	58–31 BCE
郅支 Zhizhi	tśit tśe†	çit çe	56–36 BCE
若鞮 Ruodi	ńɑk te†	nyak te	title element
復株累若鞮 Fuzhulei Ruodi	buk† ṭio lui† ńɑk te†	buk tio lui nyak te	31–20 BCE
搜諧若鞮 Souxie Ruodi	so† gɛi ńɑk te†	so gei nyak te	20–12 BCE
車牙若鞮 Cheya Ruodi	tśʰa† ŋa ńɑk te†	ça nga nyak te	12–8 BCE
烏珠留若鞮 Wuzhuliu Ruodi	ʔɑ tśo liu ńɑk te†	’a ço liu nyak te	8 BCE–13 CE
烏累若鞮 Wulei Ruodi	ʔɑ lui† ńɑk te†	’a lui nyak te	13 CE–18 CE
呼都而尸道皋若鞮 Huduershi Daogao Ruodi	hɑ† tɑ ńə† śi dou^{B†} kou ńɑk te†	ha ta nyı şi dou kou nyak te	18 CE–46 CE
攣鞮 Luandi	lyan te†	lyan te	—
虛連題 Xulianti	kʰɨɑ lian^{B†} de†	kıa lian de	—

Twenty-one *chanyu* from Touman to Punu, plus both forms of the ruling clan name: 攣鞮 in the *Shiji* and *Hanshu*, 虛連題 in the *Hou Hanshu*.

A.2 Titles and lexical items

TERM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	WHAT KIND OF ITEM IT IS: TITLE, LEXICAL WORD, OR ETHNONYM
單于 chanyu	dźan† wɑ	can va	title
撐犁孤塗單于 chengli gutu chanyu	ḍaŋ lei† kuɑ dɑ dźan† wɑ	dang lei kua da can va	title
撐犁 chengli	ḍaŋ lei†	dang lei	lexical

TERM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	WHAT KIND OF ITEM IT IS: TITLE, LEXICAL WORD, OR ETHNONYM
孤塗 gutu	kuɑ dɑ	kua da	lexical
閼氏 yanzhi / ezhi	ʔɑt dźe^{B†}	’at ce	title
屠耆 tuqi	dɑ gɨ†	da gı	title
谷蠡 luli / guli	kok† le^{B†}	kok le	title
當戶 danghu	tɑŋ† gɑ^{B}	tang ga	title
骨都侯 guduhou	kuət tɑ go	kuıt ta go	title
且渠 juqu	tsɑ^{B†} gɨɑ	tsa gıa	title
徑路 jinglu	keŋ^{C} lɑ^{C}	keng la	lexical
服匿 funi	buk† ṇɨk	buk nık	lexical
甌脫 outuo	ʔo t^{h}uɑt	’o tuat	lexical
胡祿 hulu	gɔ† lok	go lok	lexical
匈奴 Xiongnu	huoŋ nɑ	huong na	ethnonym
羯 Jie	kɨat	kıat	ethnonym

The lexical items are the only Xiongnu words the Chinese sources gloss. **撐犁** is glossed "heaven" and **孤塗** "son", which is why that pair has carried so much of the argument. The proposals for **孤塗** do not settle it. Pulleyblank compared it with Arin *bīkjàl* "son"; Vovin holds that form to be isolated within Yeniseian, the rest of the family reflecting Proto-Yeniseian **puʔ-b* "son", and finds no Altaic alternative either.[27,28] The Turkic proposal, *qut* "fortune", fits the sound and not the gloss. So the one item where a Chinese gloss and a Yeniseian etymology were thought to meet is contested by the leading advocate of that hypothesis.

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